INCH-POUND

MIL-C-48423A (AR)
AMENDMENT 3
25 September 1992
SUPERSEDING
AMENDMENT 2
30 October 1987

MILITARY SPECIFICATION

CASE, CARTRIDGE, 105MM, M14B4

This amendment forms a part of Military Specification MIL-C-48423A (AR), dated 29 May 1979, and is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1 Under Specifications, Military, Add:

"MIL-A-2550 - Ammunition, General Specification for"

Under Standards, Military, Add:

"MIL-STD-105 - Sampling Procedures and Tables For Inspection by Attributes
MIL-STD-171 - Finishing of Metal and Wood Surfaces"

Boxed in area at bottom of page, Delete:

"Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QA, Dover, New Jersey 07801" and Substitute:

"Commander, US Army Armament Munitions and Chemical Command, ATTN: SCMAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000".

PAGE 2

2.1, Under Drawings,

Delete: "US Army Armament Research and Development Command (ARRADCOM)" and Substitute:

"US Army Armament Research, Development and Engineering Center (ARDEC)".

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Delete: "8647259 - Chamber Gage" and Substitute:

"8647258 - Chamber Gage
8647259 - Acceptance Check Gage"

* Delete: "8595386" and Substitute: "12938083".

* 2.2 Under ASTM, add:

"D610 - Method for Evaluating Degree of Rusting on Painted Steel Surfaces.

* Add new paragraph 2.3 as follows:

"2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained."

PAGE 3

3.7, Line 5, Delete:

"Excess, Excess Pressure Phase (For Information Only)".

* 3.7.2, line 2, Delete: "except as permitted by Table II".

PAGE 4

* 3.8.3, Delete in its entirety and Substitute:

"3.8.3 Body/Cup/Base/Collar interface. There shall be no evidence of interference between the collar and body/cup in the region beyond the body seat groove."

* 4.3.1, Line 6, Delete the table in its entirety and Substitute:

"

PART DESCRIPTION	DRAWING	QUANTITY
Cup (uncoated)	12937901	4
Cup (coated)	12937901	10
Collar (uncoated)	12938081	4
Collar (coated)	12938081	10
Base (uncoated)	12938082	4
Base (coated)	12938082	10
Test Panels	See MIL-C-46487	2
Cartridge Case Body Blank	--------	4
Cartridge Case	12938083	54"

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PAGE 5

* 4.3.1, line 1, Change: "250 to "50".

4.3.2 Add "(see MIL-A-48078)" to the end of the paragraph.

PAGE 6

* TABLE I, Delete in its entirety and Substitute attached new TABLE I.

PAGE 7

* TABLE I, Delete in its entirety and Substitute the attached new TABLE I.

PAGE 8

4.4.3 Delete in its entirety and Substitute:

"4.4.3 Tool control dimensions.

a. Dimension marked tool control shall be gaged at the beginning or end of each working day (24 hrs), during First Article Inspection and for the first component after each tooling change, adjustment or repair. Where destruction of the component is necessary to gage such dimensions, measurements of the tool may be substituted providing the contractor has established correlation between the tool dimensions and the component dimension prior to the start of production.

b. When the contractor has satisfied (a) above, visual inspection of the tool control dimension(s) will be permitted. Visual inspection shall be performed every two (2) hours for each machine in operation. If any tool control dimension(s) is in doubt during visual inspection, the dimension(s) shall be immediately gaged to determine acceptability. Corrective action will immediately be instituted by the contractor whenever a defective tool control dimension is found."

PAGE 9

* 4.4.4.1, Delete in its entirety and Substitute attached new paragraph 4.4.4.1.

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TABLE I. - First article inspection

MIL-C-48423A (AR)
AMENDMENT 3

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Case, Cartridge, 105MM, M14B4	1 OF 2	See Below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cup, Uncoated (Dwg. 12937901)				
	Hardness	4		3.4	4.5.1
	Cup, Coated (Dwg. 12937901)				
	Examination for defects	10		3.1	4.4.4.6
	Thickness of coating	4		3.2	Gage
	Salt spray	2		3.1	4.5.3.1
	Collar, Uncoated (Dwg. 12938081)				
	Hardness	4		3.4	4.5.1
	Collar, Coated (Dwg. 12938081)				
	Examination for defects	10		3.1	4.4.4.1
	Thickness of coating	4		3.2	Gage
	Salt spray	2		3.1	4.5.3.1
	Base, Uncoated (Dwg. 12938082)				
	Hardness	4		3.4	4.5.1
	Base, Coated (Dwg. 12938082)				
	Examination for defects	10		3.1	4.4.4.2
	Thickness of coating	4		3.2	Gage
	Salt spray	2		3.1	4.5.3.1

NOTES:

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TABLE I. -- First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48423A (AR)
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PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Case, Cartridge, 105MM, M14B4				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Test Panels (See MIL-C-46487)				
	Weight of coating	2		3.1	See MIL-C-46487
	Flexibility test	2		3.1	See MIL-C-46487
	Cartridge Case Body Blank				
	Hardness	4		3.4	4.5.1
	Thickness of coating	4		3.2	Gage
	Salt spray	2		3.1	4.5.3.1
	Cartridge Case (Dwg. 12938083)				
	Examination for defects	54		3.1	4.4.4.3
	Examination of Body/Cup/Base/Collar interface	2		3.8.3	4.5.5
	Salt spray	2		3.1	4.5.3.1
	Ballistic test				
	a. Excess pressure	30		3.7	4.5.4
	b. Cold temperature	20		3.7	4.5.4

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-OA (D) Form 160, 1 Aug 83, which may not be used

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48423A (AR)
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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.4.1	Collar	1 OF 1	12938081
			NEXT HIGHER ASSEMBLY: 12938083

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None define				
Major					
101	Small outside diameter (boss dia.)		0.40%	3.1	Gage
102	Width of boss		0.40%	3.1	Gage
103	Depth of boss groove		0.40%	3.1	Gage
104	Large outside diameter (flange dia.)		0.40%	3.1	Gage
105	Circumferential taper angle of boss		0.40%	3.1	Gage
106	Circumferential taper depth of boss		0.40%	3.1	Gage
107	True position of large outside dia.		0.40%	3.1	Gage
108	Large radius of flange		0.40%	3.1	Gage
109	Protective coating missing or inadequate		0.40%	3.1	Visual
110	Surface finish		0.40%	3.1	Visual
111	Thickness of coating		0.40%	3.2	Gage
112	Salt spray	(a)		3.1	4.5.3.1
Minor					
201	Small radius of flange		0.65%	3.1	Gage
202	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

a) see 4.4.5.1

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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PAGE 10

* 4.4.4.2, TITLE, Delete: ", Prior to Coating (as applicable)".

DRAWING NUMBER, Delete: "8595383" and Substitute: "12938082".

NEXT HIGHER ASSEMBLY, Delete: "8595386" and Substitute: "12938083".

Major Category, Add major defects 106 and 107 as follows:

"106	Thickness or coating		0.40%	3.2	Gage
107	Salt spray	(a)		3.1	4.5.3.1".

Notes, Add note (a) as follows:

"(a) See 4.4.5.1."

PAGE 11

* 4.4.4.2, TITLE, Delete: ", Prior to Coating".

DRAWING NUMBER, Delete: "8595383" and Substitute: "12938082".

NEXT HIGHER ASSEMBLY, Delete: "8595386" and Substitute: "12938083".

PAGE 12

* 4.4.4.2.1, Delete in its entirety.

PAGE 13

* 4.4.4.2.1, Delete in its entirety.

PAGE 14

* 4.4.4.2.2, Delete in its entirety.

PAGE 15

* 4.4.4.2.2, Delete in its entirety.

PAGE 16

* 4.4.4.3, DRAWING NUMBER, Delete: "8595386-1-2-3" and Substitute: "12938083".

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Major defect 111, after "Body/", Add: "Cup/".

Major Category, Add major defects 112, 113, and 114 as follows:

"112	Thickness of coating	0.40%	3.2	Gage
113	Salt spray (body blank) (b)		3.1	4.5.3.1
114	Salt spray (assembly) (b)		3.1	4.5.3.1"

Notes, Add note (b) as follows:

"(b) See 4.4.5.1".

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* 4.4.4.3, drawing number, Delete: "8595386-1-2-3" and Substitute: "12938083".

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* 4.4.4, Add the attached new paragraph 4.4.4.6.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48423A (AR)
AMENDMENT 3

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.4.6	Cup	1 OF 1	12937901
			NEXT HIGHER ASSEMBLY 12938083

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None define				
Major					
101	Outside diameter		0.40%	3.1	Gage
102	Inside diameter		0.40%	3.1	Gage
103	Length, total		0.40%	3.1	Gage
104	Diameter of recess		0.40%	3.1	Gage
105	Angle of recess		0.40%	3.1	Gage
106	Thickness		0.40%	3.1	Gage
107	True position of I.D		0.40%	3.1	Gage
108	Radius at corner		0.40%	3.1	Gage
109	Surface finish		0.40%	3.1	Gage
110	Protective coating missing		0.40%	3.1	Visual
111	Coating thickness		0.40%	3.2	Gage
112	Salt spray	(a)		3.2	4.5.3.1
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

a) see 4.4.5.1

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* 4.4.5.1, Delete in its entirety and Substitute:

"4.4.5.1 Salt spray. Five samples of each component, five 4"x6" test panel samples with a minimum of one sample per coil, and five cartridge case assemblies shall be randomly selected for testing from each production lot of 25,000 or less. When lots exceed 25,000 cartridge cases, one additional sample of each component, one additional test panel, and one additional assembly shall be randomly selected and tested for every 5,000 or fraction thereof above 25,000. Samples shall be evaluated at the end of the test period in accordance with the detail specification for the protective finish. The cartridge case assemblies subjected to the salt spray test shall show no more than 1/8 inch creepage, blistering, or loss of adhesion of the coating from the scribe mark. At all other points, there shall be no more than a trace of film failure (rust grade 9 ASTM D-610), and not more than 5 scattered blisters none larger than 1mm (3/64 inch) in diameter on a 48 square inch area. Rust within one thirty second (1/32 inch) of all other edges shall be permitted. Failure to comply shall be cause for rejection of the sample. Rejection of one sample shall be cause for rejection of the lot."

* 4.4.5.2.1, Delete and Substitute:

"4.4.5.2.1 Production lot sample. Thirty (30) cartridge cases from each production lot shall be ballistically tested in the excess pressure phase. The lot shall be accepted on zero defects and rejected on one or more defects listed in Table II.

TABLE II

Defect	Accept	Reject
1. Obturation failure	0	1
2. Ejection failure	0	1
3. Primer retention failure	0	1
4. Splits or cracks in sidewall	0	1
5. Loose assembly	0	1
6. Splits or cracks within one caliber of the head that do not extend through all layers of metal	0	1"

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* 4.4.5.4, line 1, after "Body/" Add: "Cup/".

line 2, after "body/" Add: "cup/".

* 4.5.1, line 2, after "body/" Add" "and cup".

line 3, after "body" Add: "and cup".

line 5, after "body" Add: "and cup".

line 14, after "piece." Add "Four hardness values shall be taken on the cylindrical surface of the cup, excluding the flange, approximately 90 degrees apart."

4.5.2, line 3, Delete: "drawing 8647259" and substitute: "drawing 8637258".

* 4.5.3, Delete in its entirety.

* 4.5.3.1, Delete in its entirety and Substitute new paragraph as follows:

"4.5.3.1 Salt spray. Salt spray test shall be performed utilizing procedures specified in ASTM B117 and in accordance with the applicable detailed specification for the protective finish, except the salt exposure periods and solution shall be in accordance with the component drawings."

PAGE 22

4.5.4, 4.5.4.1, 4.5.4.1.1, 4.5.4.1.2, 4.5.4.1.3, 4.5.4.2, 4.5.4.2.1, 4.5.4.2.2, and 4.5.4.2.3, Delete in their entirety and Substitute:

"4.5.4 Functional proving ground. (See paragraph 6.9)

4.5.4.1 Excess pressure phase. These tests shall be conducted at a Government proving ground in accordance with applicable acceptance test procedure. The cartridge case shall be assembled into a complete round in the same manner as required for service firing. The inert loaded dummy fuzed projectile as fired weight shall be 33 $\pm$ 0.2 pounds. The sample cartridges shall be fired from an M119 Howitzer, using the PXR200-1 propelling charge. The complete round will be temperature conditioned at plus 70 degrees F $\pm$ 2.5 degrees for a minimum of twenty-four hours prior to testing. The weapon will be operated in the service manner.

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4.5.4.2 Cold temperature phase. The cartridge case shall be assembled into a complete round in the same manner as required for service firing. The inert loaded, dummy fuzed projectile as fired weight shall be 33 ± 0.2 pounds. The complete round will be temperature conditioned at minus 35 degrees F to minus 45 degrees F for a minimum of twenty-four hours prior to testing and fired in the M119 Howitzer, using the PXR200-1 propelling charge. The weapon will be operated in the service manner."

PAGE 23

* 4.5.5, Line 1, after "Body/", Add: "Cup/".

Lines 5-6, Delete: "a single ply of the body" and Substitute: "edge of body plies".

PAGE 24

* 6.6 Delete: "US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801" and Substitute the following:

"US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-QAR-I, Picatinny Arsenal, New Jersey 07806-5000."

6.7 Delete in its entirety and Substitute the following:

"6.7 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenalock Island Arsenal, Picatinny Arsenal, US Army Research and Development Command (ARRADCOM) or US Army Armament Research and Development Center (ARDC) drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC."

* 6.8 Delete in its entirety and Substitute the following:

"6.8 Submission of test data and ammunition data cards. In addition to the normal distribution of records when the cartridge case is procured by the Army Armament Munitions and Chemical Command, one copy of all ballistic test data and ammunition data cards shall be forwarded to each of the following: Commander, ARDEC, ATTN: SMCAR-CCH-P and SMCAR-QAR-Q, Picatinny Arsenal, New Jersey 07806-5000 and Commander, AMCCOM ATTN: AMSMC-QAD, Rock Island, IL 61299-6000."

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6.9 Delete in its entirety and Substitute the following:

6.9 M14B4 cartridge case. The M14B4 cartridge case has been designed, tested and type classified to operate at the pressure levels of paragraph 4.5.4.1. These pressure levels are required for use with extended range, high pressure gun systems, i.e., M119, which are currently available".

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations.idders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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